

Audit data preparation

25% faster, with more time for control testing

Our engagements were spending too much time collecting, cleaning and reconciling client data before any real audit work started. I built n8n workflows and Power BI models that do most of that preparation automatically, so the team can spend the time on testing.

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Used on	My role	Tools	Result
EY Risk & Compliance engagements	Designed and built the workflows	n8n · Power BI · Excel	25% faster data preparation

25%

faster data pull and reconciliation

2–3

engagements using the workflows so far

100%

of transactions tested, instead of samples, where the data allows

01 · THE PROBLEM

Too much of the budget went on data preparation

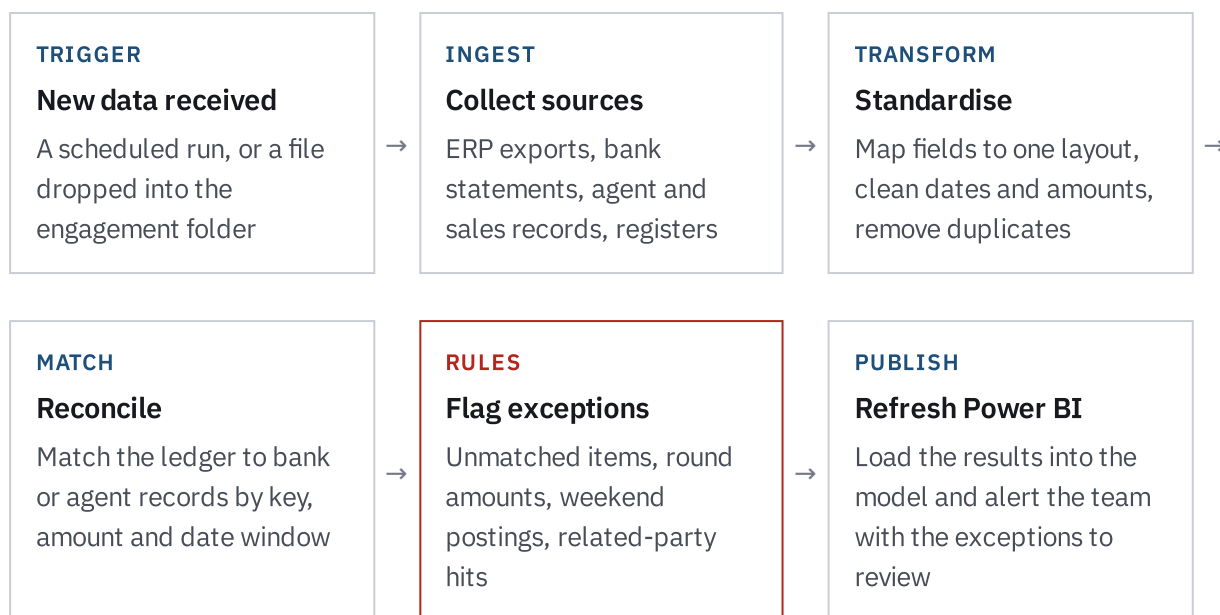
On forensic and controls engagements, data arrives in many forms: bank statements as PDFs, ERP exports, agent sales files, collection registers and emailed spreadsheets. Analysts spent days collecting, reformatting and matching it by hand. The work was slow, error-prone and different on every engagement, and it cut into the time left for control testing.

02 · EXHIBIT A

The n8n pipeline

Reconciliation workflow, from source to exception list

Representative design.



Here is how it worked on the aviation engagements. The workflow matched counter ticket sales against agent records and matched cash collections against the books. The exceptions it produced, such as collections never recorded and excess baggage not charged, became the starting point for control testing instead of something the team found at the end.

03 · TRY IT

A reconciliation, run in your browser

This is a small version of the matching and exception rules from the workflow. It uses sample data, not client records. Turn rules on or off to see how the exception list changes.

Collections: books against bank, March 2025

Sample data for demonstration

9 items tested 3 matched 6 exceptions

DATE	REFERENCE	BOOKS ₹	BANK ₹	RESULT
Mon, 03 Mar	Agent A collection	48,250	48,250	Matched
Tue, 04 Mar	Agent B collection	1,20,000	1,20,000	Round amount
Wed, 05 Mar	Counter sales	36,780	35,780	Differs by ₹1,000
Sat, 08 Mar	Voucher V-114	25,500	25,500	Weekend posting
Mon, 10 Mar	Agent C collection	62,410	—	Not in bank
Tue, 11 Mar	Agent A collection	51,090	51,090	Matched
Wed, 12 Mar	Refund R-22	50,000	50,000	Round amount
Thu, 13 Mar	Unidentified receipt	—	18,600	Not in books
Fri, 14 Mar	Agent D collection	44,375	44,375	Matched

04 · EXHIBIT B

The Power BI data model and dashboards

Star schema behind the dashboards

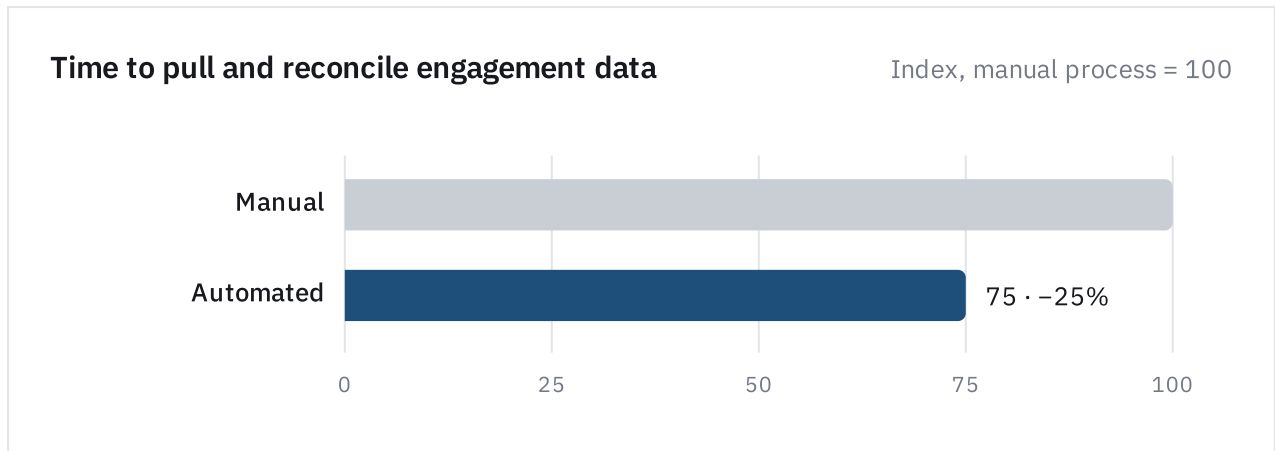
Representative.

Dim_Entity Legal entities, directors, shared addresses	Fact_Transactions Every ledger and bank line, with a match status and exception flags	Dim_Counterparty Vendors, agents and customers, with related-party tags
Dim_Date Period, fiscal year, weekend and holiday flags	Fact_Exceptions Rule hits, reviewer, disposition and evidence reference	Dim_Control Control ID, owner, COSO component, test result

The dashboards had three views. An engagement overview showed reconciliation coverage and open exceptions. An exception register let reviewers filter by rule, entity and value and record what they decided. A controls view linked each exception to the control that should have prevented it, so the audit report could go straight from the data to the root cause.

05 • RESULTS

Faster preparation, wider coverage



- **Productivity:** preparing engagement data took 25% less time, and that time went into control testing.
- **Assurance:** reconciliations cover full populations, so we test every transaction instead of a sample and exceptions can't fall between samples.
- **Consistency:** the same rules run on every engagement, and each exception links to its evidence reference for review.

06 • ADOPTION

How the team started using it

1 Start with one real engagement

Built the first workflow on a live engagement, so the time saved was measured on real work.

2 Show managers the exceptions

Walked managers through the exception register and how each flag ties back to a control finding.

3 Write it down

Wrote notes on each workflow's inputs, rules and outputs so other analysts could reuse it without me.

4 **Improve from feedback**

Added and adjusted rules based on reviewers' decisions, which reduced false positives over time.

RELEVANCE TO UAE ROLES

How this carries over to the UAE

UAE internal audit functions in banking, aviation and government are investing in continuous auditing and data analytics, following the IIA's Global Internal Audit Standards. I can build this kind of automation from day one, using whichever low-code or BI tools the team already runs.